GLENCORE

Data Standard
Equipment

12

Document properties

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1. Introduction

1.1 Master Data Standards - Purpose

The suite of Master Data Standards provide clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 Equipment Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for a SAP Equipment Master record at Glencore.
- Ensure alignment with the Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Element 5
- Identify the minimum requirements for creating, maintaining, and retiring an Equipment Master
- Provide the guiding principles to manage Equipment Master Data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is targeted at the following audience:

- Master data maintenance teams in order to understand how the object needs to be maintained
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to: Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard
- Transition to operations teams to assist in the development of user guides

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the governance for all requirements described in this Standard as well as approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders

2. Data Object Description

2.1 Definition

An Equipment is an individual, physical object that is maintained independently and is installed as part of the technical object structure. Each item of Equipment in the SAP Plant Maintenance System is defined and managed in a separate master record and has an individual maintenance history recorded against it.

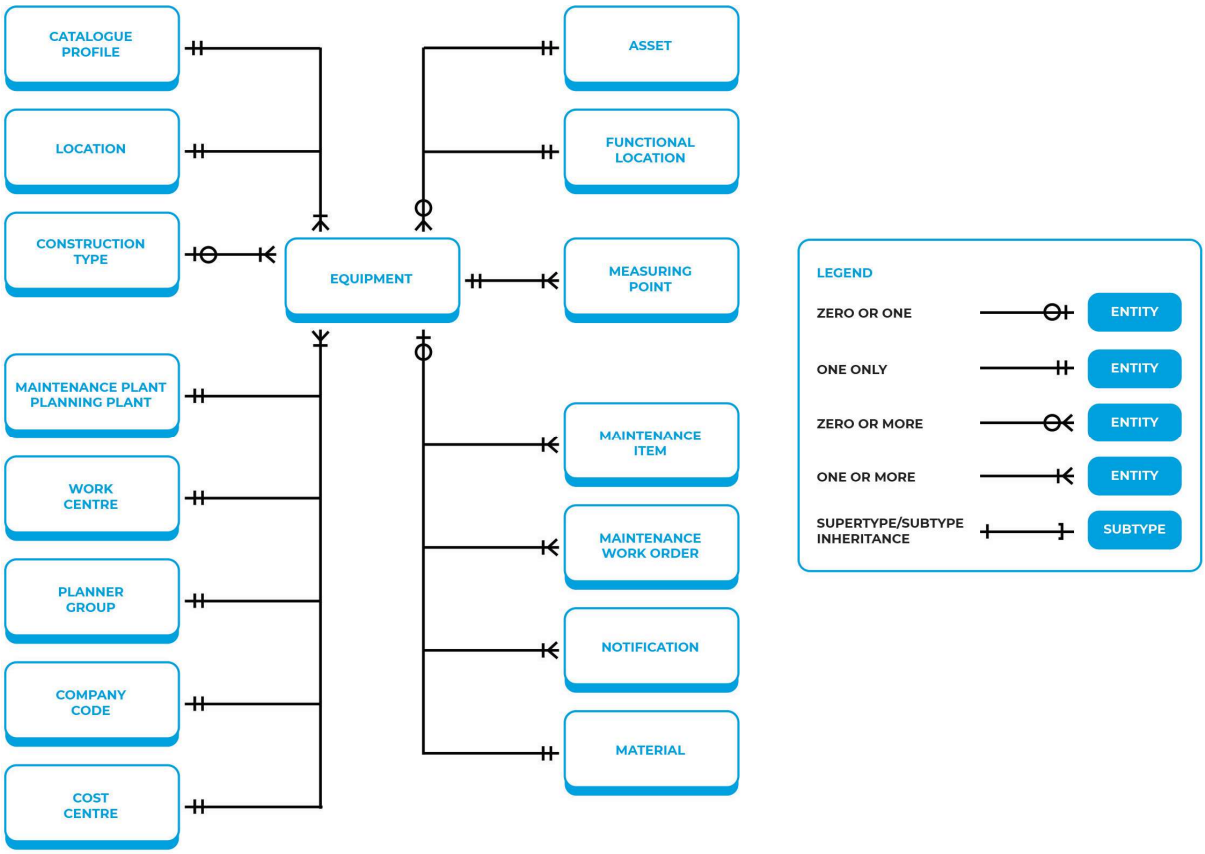
The Equipment Master record contains several types of data that uniquely identifies an individual physical/virtual object to be maintained as an autonomous unit. It is created for a Maintenance Significant Asset installed at a Functional Location (FLOC).

An Equipment Master enables you to do the following:

- Organize and track the maintenance, repair, and cost history of an asset
- Track the performance history an asset

General data	This is fixed data, which generally does not change over time. For example, the acquisition value of the item of Equipment, its size and dimensions and the year of construction.
Plant Maintenance and location data	This is time-dependent data that can change repeatedly over the course of time. This data can be, for example, the maintenance planner group, the responsible work centre, the maintenance plant and the cost centre. Time-dependent data allows you to look at an item of Equipment as it was for any particular time period. A new 'time segment' is automatically created for specific master record changes that describes the Equipment usage period.

2.2 Entity Relationship Diagram (ERD)



3. Business Requirements

3.1 Business Use

The following operational and exploration physical assets will be represented by Equipment:

- Glencore owned assets for which there is a maintenance strategy, including:
    - Assets maintained by a 3rd Party.
    - Tracable equipment components.
- Glencore maintained assets, even if Glencore does not own the asset. This would also include hired assets.

Equipment will not be created for Physical assets owned and maintained by Service contractors.

Equipment Category

The Equipment Category allows you to control:

- Number interval and number assignment.
- Activation of change documents.
- Assigning a view profile.

The following equipment category will be used

- M - Machines
- V – Vehicle

Sub Equipment

Sub-Equipment are subsets of equipment that have a Equipment parent rather than a functional location. Sub-equipment can be to a maximum of 3 levels.

Rotable Equipment

Equipment will also be used to identify Rotable Equipment.

Rotable Equipment is managed using both an Equipment Master in Plant Maintenance, and a Material Master within the framework of inventory management. This comprises:

- A unique combination of material and technical identification number/serial number (optional)
- Stock information
- Warehouse information

3.2 Business Design Rules & Criteria

The following are key points to building an Equipment technical object structure:

- The Equipment technical object structure will be used to define the maintenance objects at each site
- Maintenance will be performed from Equipment level down & defines the Maintainable Unit / Item(s) as the sub structure
- Existing tag identification information will be stored against the Equipment Sort field & Tech ID field, and will be available in searching and reporting. This will minimise the system migration change impact and allow continual cross-referencing to drawings, asset identifiers, other system tags, etc
- The number of levels used in the structure will be determined by the complexity of the asset
- Maintainable Unit Equipment will have Classification and Characteristic attributes to enable integration and reporting (this will also determine part of the equipment description)

3.2.1 Equipment Master Data Life Cycle

The following diagram illustrates the Equipment Master Data Life Cycle process:



When **creating** an Equipment Master, the following specific requirements must be met. An Equipment Master:

- shall be created for each Maintenance Significant Asset
- shall be created for each replacement of a serialized Maintenance Significant Asset, where applicable
- shall be installed to a relevant Functional Location (FLOC)
- shall have an Asset Tag or Tech ID to identify it
- shall be categorized and described by the following:
    - Category
    - Object Type
    - Class
    - Description
- shall contain all relevant technical information to allow efficient operational use
- shall have data field inheritance maintained
- should have manufacturer details and be assigned a unique and valid serial number
- shall be linked to a Material Master

When **maintaining** an Equipment Master, the following specific requirements must be met. An Equipment Master:

- for Rotable equipment, shall have movement tracking of the individual Material maintained in the Facility throughout its life cycle (e.g. In/out of Warehouse, in/out of Vendor/Onsite repair shop for refurbishment/repair. See Rotable Equipment)
- shall have data field inheritance information throughout its lifecycle maintained

When **retiring** an Equipment Master, the following specific requirements must be met. When retiring an Equipment Master the end user:

- shall gain approval from a designated approver or approvers
- shall follow a the Equipment deletion process (see Retiring Equipment)

3.2.2 Rotable Equipment

Rotable Equipment must meet all the following requirements:

- It can be repeatedly installed and removed from operation and economically restored to a fully serviceable condition.
- Installation history and tracking is required.
- A link to an inventory catalogue code (material number) and inventory tracking is required.
- A material number and a unique Tech ID/serial number for life of asset tracking are required in both inventory management and operations. The serial number (optional) assigned to the Equipment record will be the same as the serial number that is recorded on the physical component.

An item of Rotable Equipment must always be in one of the following four states:

- Installed on Equipment.
- Under repair in a refurbishment order (order type PM04).
- Under Inventory control including: damaged items awaiting repair; new and repaired items in the store; damaged items being repaired offsite.
- Marked for deletion (scrap).

3.2.3 Financial Asset linkage

The Equipment Maintenance Cost Centre will be linked to the equivalent Equipment Operating Cost Centre.

3.3 Hierarchy Requirements

Equipment must be installed in a Functional Location when they are in productive use to form part of the Technical Object Hierarchy.

Equipment must be installed at level 5 of the Functional Location Structure.

There may be up to 4 equipment levels including sub equipment

Rotable equipment when not installed to productive equipment will be under Inventory control and not be part of a hierarchy.

3.3.1 Retiring Equipment

Equipment that is to be retired/decommissioned or disposed of will be detached from it's operational position in the Technical Object Hierarchy and attached to a Graveyard Functional Location that is local to each Operation. The agreed mask is defined as follows and mirrors the mask definition provided in the Functional Location Master Data Standard :

- XXXX-DEL-DEL-DEL-EQPDEL for each Operation (XXXX)

4. Detail Standard

4.1 Data Object Coding Standard

Each Equipment number will be a system-generated number, unique across the organisation. The Sort field & Tech ID will be used to identify the Equipment colloquially according to the site's tagging system.

4.1.1 Equipment

Application Definition	Equipment Number
Application Guiding Principles	Number by which an item of Equipment can be clearly identified.
Guiding Principles	Will be internally system generated. Physical tag id of the Equipment will be stored in the Sort field & Tech ID.
Application Coding Convention	18 characters numeric
Naming Convention	Refer to Section 4.1.

4.1.2 Equipment Short Text

Application Definition	Equipment Short Text
Application Guiding Principles	Description is in the form of short text.
Guiding Principles	Will describe the Asset
Application Coding Convention	40 Characters Alphanumeric
Naming Convention	Description is derived from the Tech ID field, Equipment Classification Name followed by Mandatory Charateristic values.

4.1.3 Equipment Construction Type

Application Definition	Equipment Construction Type
Application Guiding Principles	Material number under which pieces of Equipment of the same construction are grouped.
Guiding Principles	Construction Type will not be the same material number as the one used to represent the Equipment in Inventory Management. It is to be a material of type IBAU. Provides a data link between the Equipment and any associated BOM and Task Lists
Application Coding Convention	18 characters numeric
Naming Convention	Internally generated number for materials.

5. Appendices

5.1 Complete Field List

- **Table –** SAP table where data object appears
- **Field –** SAP data attribute field
- **Field Name –** SAP field name of data object
- **Description –** Description of input required
- **Data Type –** SAP data type (CHAR=Character, DATS=Date, CUKY=Currency key, UNIT=Measurement type, NUMC=Number)
- **Length –** Character length of field
- **MOCD –** Data requirement (M=Mandatory, O=Optional, C=Conditionally mandatory, D=Data Origin i.e. Inherited from the Superior Functional Location)
- **Field Validation Type –** Entry type allowed (e.g. Free text, List selection) and additional information

Table	Field	Field Name	Description	Data Type	Length	M/O/C/D	Field Validation Type
ITOBATTR	EQUNR	Equipment number	Number by which a piece of equipment can be clearly identified	CHAR	18	M	System generated which defaults after creation
ITOB	DATAB	Date valid from	Date indicating as of when an entry is valid	DATS	10	M	Valid date format: DD.MM.YYYY
ITOB	EQTYP	Equipment category	Key for distinguishing individual technical objects according to their use	CHAR	1	M	M – Machines V – Vehicles

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Table	Field	Field Name	Description	Data Type	Length	M/O/C/D	Field Validation Type
ITOB	SHTXT	Description of technical object	Description of Equipment	CHAR	40	M	Description is derived from the Tech ID field, Equipment Classification Name followed by Mandatory Charateristic values.
ITOBATTR	KLASSE	Class Number	Name used to uniquely identify a class within a class type	CHAR	18	M	Table selection
ITOB	EQART	Object Type	Key that describes the type of technical object. Select from list	CHAR	10	M	Table selection
ITOB	GROES	Size/dimension	Size/dimension of the object	CHAR	18	O	Non validated free text field
ITOB	BRGEW	Weight	Numeric entry of weight of object	CHAR	17	O	Non validated free text field.
ITOB	GEWEI	Weight Unit	Units of weight – kg, tonnes etc.	UNIT	3	C	Mandatory when weight specified BRGEW
ITOB	INBDT	Start-up date	Time at which the Equipment came into operation.	DATS	8	O	Validate that date is in the Format DD.MM.YYYY
ITOB	ANSDT	Acquisition date	Date purchased	DATS	10	C	Validated date field DD.MM.YYYY.
ITOB	ANSWT	Acquisition Value	Purchased cost	CHAR	17	C	Non validated free text field.
ITOB	HERST	Manufacturer	Manufacturer of asset	CHAR	30	C	Validated free text field (MDG).

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Table	Field	Field Name	Description	Data Type	Length	M/O/C/D	Field Validation Type
ITOB	HERLD	Manufacture Country	Country of manufacture	CHAR	3	O	Valid country key from table.
ITOB	BAUJJ	Construction Year	Year in which the equipment was constructed	CHAR	4	O	Non-validated free numeric field NNNN.
ITOB	BAUMM	Construction Month	Month of the year in which the equipment was constructed	CHAR	2	O	Table select.
ITOB	TYPBZ	Model number	Manufacturer model number	CHAR	20	C	Table select: Validated against the manufacturer table.
ITOB	MAPAR	Manufacturer part number	Manufacturer part number	CHAR	30	O	Non-validated free text field.
ITOB	SERGE	Manufacturer serial number	Manufacturer serial number	CHAR	30	O	Non-validated free text field.
ITOB	BEBER	Plant section	Number which clearly identifies a plant section.	CHAR	3	N	Table select: value from plant section table.
ITOB	ABCKZ	ABC indicator	Criticality/risk category of Equipment.	CHAR	1	M	A – Critical B- Important C – Normal N – Not Maintainable

Table	Field	Field Name	Description	Data Type	Length	M/O/C/D	Field Validation Type
ITOB	EQFNR	Sort field	This field contains a company-specific sort term for the equipment, generally the tag number	CHAR	30	M	Non-validated free text field containing the physical asset tag number. Derived from: • Plant – 3 Character Alpha • Area – 3 Character Numeric • EquipmentType – 2 Character Aplha • Unique Identifier – 22 Character Aplha Numeric
ITOB	BUKRS	Company Code	The company code is an organisational unit within financial accounting.	CHAR	4	M	Defaults from maintenance plant – no entry required.
ITOB	KOSTL	Cost Centre	Key uniquely identifying a cost centre – used for OPEX and becomes the default work order settlement object.	CHAR	10	C	Table Selection: Must be an Equipment Maintenance cost centre.
ITOB	IWERK	Maintenance Planning Plant	Number that clearly identifies a maintenance plant.	CHAR	4	M	5030 – Cobar 7490 – Ernest Henry 8110 – Mount Isa Mine 8111 – Townsville Port 9830 – Copper Refinery
ITOB	INGRP	Planner Group	Key that uniquely identifies the planner group in Plant Maintenance.	CHAR	3	C	Table select valid planner group.

Table	Field	Field Name	Description	Data Type	Length	M/O/C/D	Field Validation Type
ITOBATTR	GEWRK	Main Work Centre	Key which uniquely identifies the responsible work centre for the equipment	CHAR	8	C	Table select valid Work Centre.
ITOBATTR	WERGW	Plant Associated with MWC	Plant in which the main work centre that has been entered is to be found.	CHAR	4	D	Will default with the selection of Plant.
ITOB	RBNR	Catalog Profile	Catalog of Object Parts and Cause and Damage codes.	CHAR	9	C	Default value is ZPM000001 – PM Standard Catalog Profile
ITOB	TPLNR	Functional Location	The Functional Location into which the Equipment will be installed.	CHAR	40	C	For all Equipment and Sub Equipment. Exception for Rotable Equipment when under Inventory control.
ITOB	HEQUI	Superordinate Equipment	Number that clearly identifies the superior Equipment associated with this piece of Equipment.	CHAR	18	C	For Sub Equipment only.
ITOB	POSNR	Position	The position of the Equipment in thenical object structure.	CHAR	4	O	Non-validated free text field.
ITOB	TIDNR	Technical identification number	Sort criterion which can be freely chosen for the technical object.	CHAR	25	M	Non-validated free text field. Shall match the Sort Field.
ITOB	SUBMT	Construction type	Material number under which pieces of Equipment of the same construction are grouped.	CHAR	18	M	Table select. See Equipment Construction Type in the Attributes section of this document.
ITOB	MATNR	Material Number	Link between Equipment number and material catalogue.	CHAR	18	O(/M)	Table selection. Mandatory for Rotable Equipment Only
ITOB	SERNR	Serial number	Specifies the serial number of a serialised material. Serial number will match the serial number of the physical component where possible.	CHAR	18	C	Mandatory for serialised Equipment Non-validated free text field.

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Table	Field	Field Name	Description	Data Type	Length	M/O/C/D	Field Validation Type
DIADR	NAME1	Name 1	Description of the location of the physical asset.	CHAR	50	O	Non-validated free text field.
DIADR	NAME2	Name 2	Additional descriptive information for the location of the physical asset.	CHAR	50	O	Non-validated free text field.
DIADR	STREET	Street	Descriptive information for the location of the physical asset – specific address.	CHAR	50	O	Non-validated free text field.
ITOB	ANLNR	Asset	This is the number which, together with the asset sub number, identifies a fixed asset in asset accounting.	CHAR	12	O	Table selection.
ITOB	ANLUN	Sub Asset	This is the number which, together with the main asset number, identifies a fixed asset in asset accounting. Provides the ability to track later acquisitions to assets.	CHAR	12	O	Table selection.

6. Stakeholder Signoff

Stakeholder Name	Role	Contact	Stakeholder Sign-off
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